

Concise Design Document

Goals and Requirements

The primary goal is to develop a real-time, scalable, and personalized movie recommendation system. The system ensures:

- **Low-Latency Recommendations:** Deliver suggestions within **100 milliseconds** using pre-trained models and server-side inference.
- **Personalized Feedback:** Uses **content-based filtering and SVD** to adjust recommendations dynamically.
- **Hybrid Model:** Runs **collaborative filtering (SVD) and content-based filtering** in parallel, selecting five random suggestions for user tastes.
- **Scalability:** Uses **Docker** and **Kubernetes** for containerized deployment and auto-scaling.
- **Security:** Passwords are hashed with **Argon2** to ensure strong protection.

Architectural Decisions

Key design choices optimize system efficiency:

- **Hybrid Model:** Integrates **SVD and content-based filtering** for diverse recommendations.
- **Microservices:** Applied only to **frontend and backend** for easier updates and scaling.
- **Server-Side Inference:** Enables seamless model updates while minimizing latency.
- **Database - MongoDB:** Chosen for **flexible schema, scalability, and efficient querying**.

Dataset & Preprocessing

The system uses the **MovieLens dataset**:

- **Movies** (titles, genres, years), **Ratings** (user interactions), **Tags** (user-generated keywords), **Links** (external metadata).

Preprocessing:

- Merges **tags with movies**, extracts **release years**, splits **genres**, and encodes them.
- Stores processed data in **MongoDB**, preventing redundant insertions.

System Overview

Machine Learning Models:

- **Content-Based Filtering:** Uses **cosine similarity** to match movie attributes.
- **Collaborative Filtering (SVD):** Identifies hidden patterns in user-item interactions.
- **Hybrid Model:** Combines both approaches and selects five random recommendations.

User Profiling & Feedback:

- Maintains **long-term and session-based profiles** for evolving personalization.
- **Real-time ratings (1–5 scale)** adjust future recommendations dynamically.

Features

- **Monitoring:** Uses **Minikube dashboard** for tracking system health.
- **Scalability:** **Kubernetes-based containers** ensure smooth scaling.
- **Adaptability:** Model iteratively adjusts based on evolving user preferences.
- **Customization:** Users can modify **genres, directors, and model selection**.